

ORIGINAL ARTICLE OPEN ACCESS

Energy Modelling and Analysis for Sustainable Energy System of Somalia

Abdisalan Moallim Hirsi¹ | Shoaib Ahmed Khatri¹ | Nayyar Hussain Mirjat¹  | Musavir Hussain² | Laveet Kumar³ | MD Shouquat Hossain⁴ 

¹Department of Electrical Engineering, Mehran University of Engineering and Technology Jamshoro, Jamshoro, Sindh, Pakistan | ²Department of Electrical Engineering, Mehran University of Engineering and Technology SZAB Khairpur Mir's, Khairpur Mir's, Sindh, Pakistan | ³Department of Mechanical Engineering, Mehran University of Engineering and Technology Jamshoro, Jamshoro, Sindh, Pakistan | ⁴Department of Electrical and Electronics Engineering, School of Engineering, International University of Business Agriculture and Technology, Dhaka, Bangladesh

Correspondence: MD Shouquat Hossain (shouquat64@gmail.com)

Received: 3 July 2025 | **Revised:** 11 October 2025 | **Accepted:** 22 October 2025

Keywords: energy systems | environment | environmental impact | policy

ABSTRACT

To ensure sustainable management of energy systems and maintain their benefits over time, it's imperative to prioritise science-based decision-making models throughout energy policy formulation and management processes, however, there is a significant research gap regarding energy modelling in Somalia. Therefore, this study developed three scenarios namely: Business as usual (BAU), Renewable Energy Technologies (RET) and visionary transition scenario (VTS) for supply side to fulfil projected demand. Considering, CO₂ emissions resource potential, and techno-economic parameters. The novelty of this study lies in being the first comprehensive application of the Long-Range Energy Alternatives Planning (LEAP) model for Somalia's electricity sector, integrating environmental, economic, and technological perspectives to support sustainable energy transition policies. Using the LEAP model, electricity demand was projected to increase from 133.2 GWh in 2020 to 1814.2 GWh in 2050, representing a growth of approximately 13.62 times compared to the base year, with an estimated annual rise of 1.88% under the BAU and other scenarios in this study. In RET scenario, where 70.3% of electricity generation comes from renewable sources, carbon dioxide (CO₂) emissions are forecasted to be significantly lower compared to the BAU scenario in the year 2050, with emissions reaching 1406.3 thousand metric tonnes which is 1.56 times smaller than the BAU scenario for the same time. CO₂ emissions in VTS scenario are estimated to be net-zero by 2050, driven by a complete shift from fossil fuels to renewable energy sources. Greenhouse gas emissions for each scenario were evaluated using the IPCC Tier 1 methodology, and the economic assessment based on Net Present Value (NPV) analysis shows that the VTS scenario offers the most cost-effective and sustainable pathway. Overall, the findings highlight that expanding solar and wind energy can enable Somalia to achieve a low-carbon and economically resilient electricity system, offering crucial insights for national policymakers and future sustainable development planning.

Abbreviations: BECO, banadir electric company; BUA, business as usual; CO₂, carbon dioxide; DG, diesel generator; EAC, East African Community; ESPs, electricity service providers; GHG, greenhouse gas emission; GWh, gigawatt hour; IEA, international energy agency; IPCC, intergovernmental panel on climate change; LAS, League Of Arab States; LEAP, Long-Range Energy Alternatives Planning System; LMDI, logarithmic mean division index; MW, megawatt hour; NAM, Non-Aligned Movement; NDP, National Development Plan; Nemo, next energy modelling system for optimisation.; NPV, Net present value; OIC, Organisation Of Islamic Cooperation; RE, renewable energy; RET, renewable energy technology; SURE, sustainable rural energy decision support system; UN, united nations; VST, visionary transition scenario.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Energy Science & Engineering* published by Society of Chemical Industry and John Wiley & Sons Ltd.

1 | Introduction

The energy crisis in Somalia reflects long-standing weaknesses in national energy planning and policy frameworks, with limited regulations and oversight as well as critical shortages of qualified personnel. This is compounded by the high cost of electricity and reliance on expensive diesel generators. All these challenges are further exacerbated by the existing insecurity and political instability within Somalia. There is also a lack of proper information among Somalia's policymakers about the possible impacts of different energy policy pathways [1]. Moreover, Somalia is ranked among the most vulnerable countries to climate change in the world, despite its carbon emissions being only 0.8% [2]. Addressing these challenges, the NDP-9 (2020–2024) outlines the Ministry of Planning's strategic framework to diversify energy supply through solar, wind, and geothermal sources [3]. However, practical progress toward this diversification has been limited due to institutional barriers, lack of investment, and insufficient technical assessment of renewable potential.

Giganti et al. [4] revealed through a Topic Modelling analysis that the relationship between institutional quality and renewable energy is multifaceted, emphasising that holistic and well-functioning institutional policies are crucial for driving sustainable energy transitions. In a related study, Giganti and Falcone [5] highlighted that the circular economy and energy transition are interlinked systems driven by institutional quality, innovation, and policy coherence, underscoring the need for integrated strategies to achieve sustainable low-carbon development. Furthermore, recent research by Falcone and Errichiello [6] emphasises that achieving sustainable energy transitions also requires addressing energy justice concerns, ensuring equitable access, affordability, and inclusivity in energy policymaking to reduce socioeconomic disparities and promote environmental well-being. Expanding the research Barra and Falcone [7] found that economic policy uncertainty can worsen environmental inefficiency, while strong institutional quality and stable political frameworks play a crucial moderating role in enhancing environmental sustainability. Falcone and Tutore [8] mapped the nexus between Transformative Innovation Policies (TIPs) and Sustainable Development Goals (SDGs) through bibliometric and social network analysis, highlighting that innovation-driven policies and multi-stakeholder collaborations are essential to accelerate sustainable transitions and achieve equitable development outcomes. Aldieri et al. [9] emphasised that achieving sustainable energy transitions also requires addressing energy justice concerns, ensuring equitable access, affordability, and inclusivity in energy policymaking to reduce socioeconomic disparities and promote environmental well-being.

The shift to a sustainable, low-carbon society necessitates evidence-based decision-making. Science-informed energy models offer a structured approach for gathering insights to inform government policies and energy management strategies. These models facilitate data collection, hypothesis testing, and outcome evaluation to identify optimal solutions within defined constraints [10]. In the global context, integrated energy modelling has become a cornerstone for designing effective energy transitions and evaluating trade-offs between economic growth, environmental sustainability, and energy security.

Numerous modelling tools have been developed to support evidence-based policy formulation in both developed and developing contexts. However, effective long-term energy planning requires more than just system modelling. Models must predict future trends and propose policies to ensure energy sustainability over time. Regrettably, Somalia currently lacks such effective policy and planning applications to guide its energy sector transition [11]. This absence of structured energy modelling frameworks limits Somalia's ability to anticipate future demand, assess renewable integration, and design low-carbon development pathways.

As depicted in Table 1 There are several energy modelling tools and techniques, and there have been several studies conducted in different countries with different time horizons. These studies demonstrate how scenario-based modelling assists policymakers in evaluating alternative energy futures and identifying sustainable pathways.

Based on the extensive review conducted by the author, no comparable studies have been conducted on energy modelling and analysis for a sustainable energy system in Somalia. Considering the identified literature gaps, this study will be the first to apply a scenario-based energy modelling approach for Somalia using the Long-Range Energy Alternatives Planning (LEAP) model from 2020 to 2050. By combining locally compiled data with international sources, it extends previous LEAP applications and provides context-specific evidence for a post-conflict, low-electrification country. This approach allows a comprehensive assessment of future electricity demand and supply dynamics under multiple policy scenarios, incorporating socioeconomic and demographic trends, renewable potential, and emission constraints. The energy demand projections will be related to socioeconomic and demographic factors. The scenarios will represent potential future pathways to aid policy and decision makers in understanding the impacts of continuing or altering these pathways. The analyses will provide details on resource utilisation, cost parameters, energy conversion processes, and carbon dioxide emissions under different cases and scenarios. This study employs LEAP, a widely adopted integrated energy modelling tool. LEAP tracks energy consumption, production, and resource extraction, encompassing energy and nonenergy sector greenhouse gas (GHG) emissions [12]. Additionally, it assesses local and regional air pollutant emissions, making it suitable for studying climate co-benefits through pollution reduction. Applying LEAP in Somalia's context is particularly significant because it enables a structured analysis despite data scarcity and institutional challenges.

Despite Somalia's abundant energy resources, the country has nevertheless experienced relatively limited energy usage, especially in electricity consumption. This paper aims to address crucial policy questions by employing the LEAP system while considering principles of sustainable energy systems, such as resource scarcity, social conditions, economic factors, and environmental impacts. The key policy questions are: What is Somalia's current energy situation and existing energy plans? What will Somalia's future energy supply pathways look like? How will each future energy supply pathway impact costs and the environment?

It is anticipated that this study will develop energy models and future energy supply pathway scenarios and analyse their techno-economic and environmental impacts. The study will

TABLE 1 | Previous studies undertaken using energy modelling tools and techniques.

Ref	Country	Year	Modelling tool	Findings Of the study
[11]	Pakistan	2018	LEAP	The Renewable Energy Technologies scenario, though capital intensive initially, was found to be the most sustainable electricity generation path followed by the Energy Efficiency and Conservation scenario which had the lowest demand and net present value. This suggests renewables with energy efficiency can provide sustainable solution. Conservation (EEC) and renewable energy technology scenario would fit most the economic and environmental needs of Pakistan.
[12]	Pakistan	2022	LEAP	Long-term energy planning with flexibility through scenario analysis provides opportunities to introduce necessary policy and infrastructure changes over time to transition towards sustainability goals.
[13]	China	2018	LEAP	Scenario analysis is important to understand the impacts of various policy measures and technological changes. The article defines 50 scenarios including a baseline and various single measure, sectoral, and system-wide compound scenarios.
[14]	China	2021	LEAP	Results show potential for emissions reductions and energy savings. Energy modelling can quantify outcomes of sustainable energy strategies for a country.
[15]	Mexico	2018	MEM70	this paper demonstrates how energy system modelling and analysis can provide useful insights and support decision-making for developing sustainable energy systems at the national level.
[16]	Germany	2021	SURE	The paper discusses the importance of energy modelling and planning tools like HOMER, LEAP and Network Planner for designing optimal decentralised energy systems, especially for rural electrification in developing countries.
[17]	Turkish	2023	LEAP	The LEAP (Long-range Energy Alternatives Planning) modelling tool can be useful to evaluate different policy scenarios and their impacts on energy use, costs, and emissions over the long-term given its integrated, scenario-based approach
[18]	South Korea	2019	LEAP	Developing multiple scenarios considering different levels of renewable energy deployment, electrification, and energy demand reduction can help illustrate different policy pathways.
[19]	Indonesia	2022	LEAP	Capacity building on use of modelling tools like LEAP may be needed to enable national experts to periodically update and refine energy projections to guide long-term policymaking.
[20]	Bangladesh	2020	LMDI	It's found that economic growth, electricity consumption, energy consumption, and population growth positively impacted CO2 emissions, while renewable energy deployment helped reduce emissions.
[21]	Nepal	2019	LEAP	Promoting renewable energy is key for meeting future energy needs sustainably. The modelling showed massive adoption of renewables must be part of Nepal's energy plans. Renewables can help reduce dependence on energy imports
[22]	Kenya	2022	LEAP WITH NEMO	The study investigated the impact of alternative electricity development pathways on system costs and greenhouse gas emissions in Kenya.
[23]	Pakistan	2021	LEAP	Developing an integrated energy modelling framework that combines both demand-side and supply-side analysis is important for a comprehensive analysis of the energy sector and to inform effective policymaking. The proposed modelling approach in the paper integrates demand forecasting with supply optimisation to better analyse the impacts of different scenarios.
[24]	Brazil	2019	LEAP	integrating both energy and urban planning policies can yield greater benefits than either approach alone
[25]	Ecuador	2021	Energy PLAN	National and local development plans and policies should inform energy modelling scenarios to ensure feasibility and alignment with broader goals. The paper bases its scenario on Ecuador's regulatory framework and development plans.

(Continues)

TABLE 1 | (Continued)

Ref	Country	Year	Modelling tool	Findings Of the study
[26]	Norway	2023	LEAP	This paper demonstrates how energy modelling and analysis using tools like LEAP can provide valuable insights and inform planning for a more sustainable energy system through developing long-term scenarios, validating models with national data, forecasting demand drivers, and evaluating mitigation policy measures - all of which are important for energy planning at the national level.
[27]	Ethiopia	2024	LEAP	Accurate energy forecasting enables power system operators to anticipate consumption patterns, make informed decisions about energy generation and distribution, and ensure a reliable and stable power supply for businesses and consumers.
[28]	Niger	2024	LEAP	Greenhouse gas emissions (CO ₂ , CH ₄ , and N ₂ O) are expected to be significantly reduced in scenarios with more renewable energy and natural gas compared to the reference scenario
[29]	Iran	2024	LEAP	The paper discusses renewable energy development in rural areas of Iran, emphasising the importance of sustainable energy systems.
This study	Somalia	2024	LEAP	It's found that a transition to renewable energy sources can significantly reduce emissions and be economically viable for Somalia's future energy system.

also devise vital long-term energy policy recommendations for the development of low-cost, low-carbon energy in Somalia and strengthen planning for future energy needs. By providing quantitative evidence for alternative policy options, this study contributes to the wider literature on sustainable energy planning and supports national strategies for cleaner and more resilient energy systems.

The rest of the paper is structured as follows: Section 2 Electricity demand-supply status and exiting policies of Somalia. Section 3 Methodology and Data. Section 4 Scenario development. Section 5 Results. Section 6 discussion and Section 7 conclusion. Section 8 lists the references.

2 | Electricity Demand-Supply Status and Exiting Policies of Somalia

Studying a nation's future energy systems should consider the energy needs across different sectors and fuel types, as well as the available energy sources that impact supply security. This involves examining current energy system trends and features, such as demand, supply, and gathering up-to-date data [26]. The electricity demand-supply status of Somalia is as follows:

The United Nations (UN) Population Division estimated the population of Somalia to be approximately 16.86 million as depicted in Figure 1, with a growth rate of 2.92% in March 2022 as Figure 2 shows. Only 35% of the Somali population resides in urban areas, with the remaining 65% living in rural regions. Rural livelihoods are largely sustained by agriculture and charcoal production, where traditional biomass fuels like firewood and charcoal are commonly utilised for income generation [24]. The total number of households in the country is 2.6 million, with an average household consisting of 6 members [3]. There are three most electricity consumer groups in Somalia: domestic users, small industrial, commercial, and street lighting consumers, and other electricity-consuming public services provided by the government. However, the absence of federal oversight has led to uneven growth in the energy sector over the last three decades. Urban centres generally have better access to electricity compared to rural areas, where access to power is severely limited. So as Figure 3 shows in 2019 only approximately 36% of Somalia's total population and 11% of its rural population had access to electricity, leaving around 9.88 million people without adequate access. This situation makes Somalia one of the least electrified countries globally [25].

As depicted in Figure 4, Somalia lies on the eastern coast of the Horn of Africa, bordered by Kenya, Ethiopia, and the Republic of Djibouti. It spans an area of 638,000 square kilometres, positioned at coordinates 10°N, 49°E. The climate is primarily hot and dry throughout most months of the year [29]. Climatically, Somalia is characterised as hot, arid to semi-arid, experiencing two main rainy seasons: the “Gu” rains from April to June and the “Deyr” rains from October to December. Rainfall is low and irregular, with the possibility of two or more consecutive years of failed rains occurring every one in 3 years. The northern coast and northeast regions are the driest areas, while the northwest and southwest are the wettest. In the southwest, rainfall surpasses the annual average. On average, Somalia receives about 250 mm of rainfall annually [32]. Somalia is divided into 18 regions covering a total

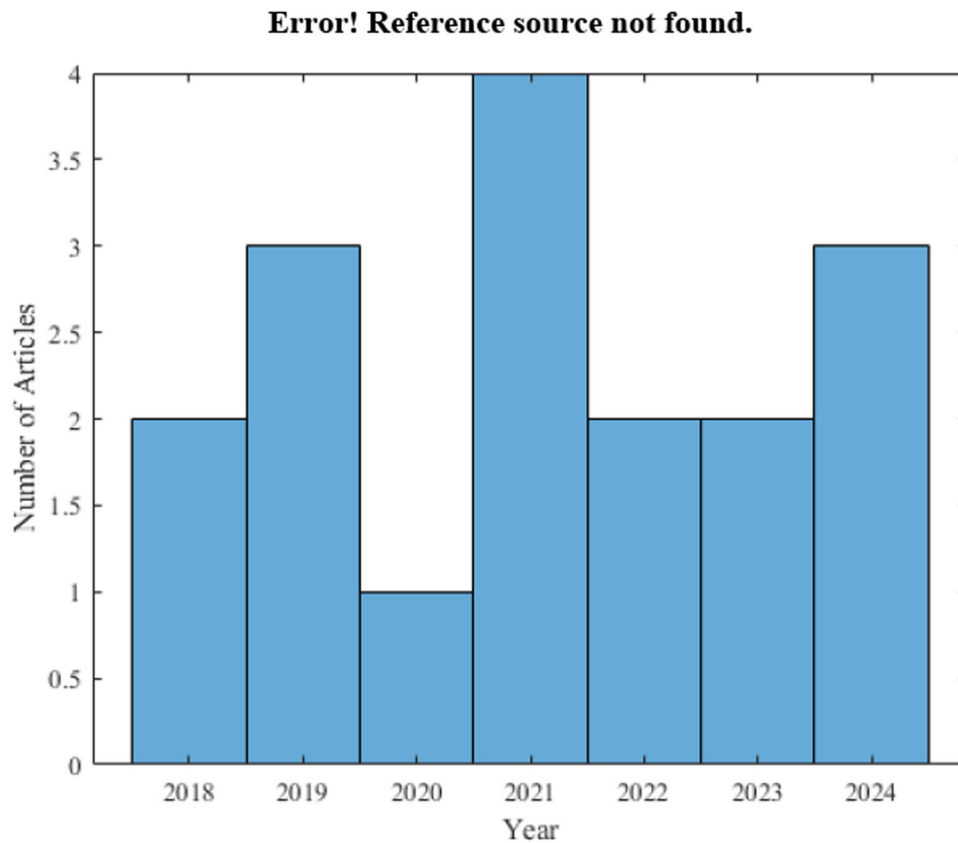


FIGURE 1 | Research trends electricity demand status.

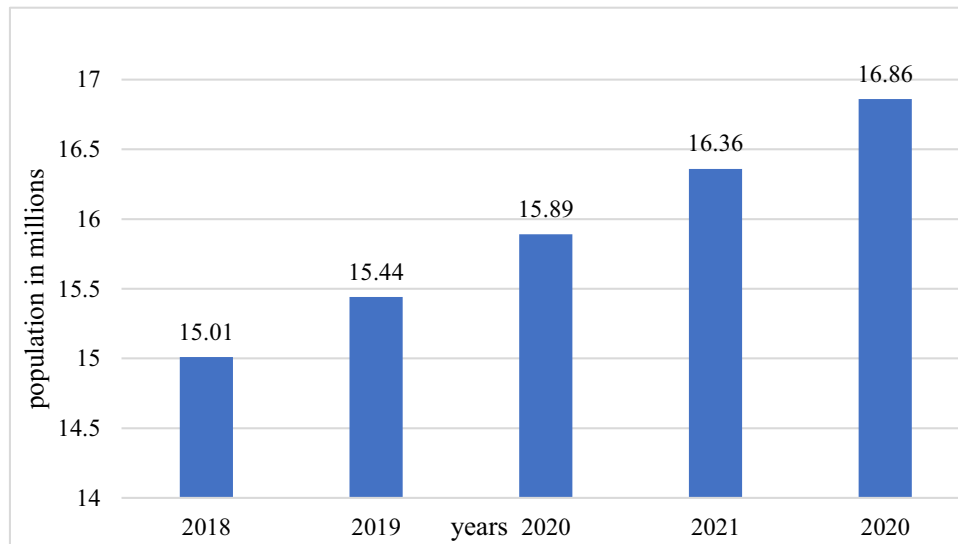


FIGURE 2 | The population in Somalia from 2018 to 2022 [1].

area of 637,540 square kilometres. Additionally, it boasts a coastline stretching over 3333 kilometres, making it mainland Africa's longest coastline. The capital city, Mogadishu, lies slightly north of the equator along the shores of the Indian Ocean [33]. Somalia's GDP in 2019 was \$5973 million, with a growth rate of 2.7%; in 2020, it was \$5,956 million, with a growth rate of -0.3% , indicating a decline attributed to COVID-19 as shown in Table 2.

Somalia is a member of these international and regional organizations. UN, EAC, LAS, NAM, OIC and other organisations [34].

2.1 | Electricity Generation Capacity Estimation

Pre-conflict, the Somalia National Electric Corporation (Ente Nazionale Energia Elettrica-ENEE) was the single public utility in operation [27]. Currently there is no national electricity grid in Somalia. The system consists of isolated island networks run by private electricity providers Each provider owns generation sources (mainly diesel generators), distribution networks, and directly serves customers within their area [35] The energy capacity of Somalia is around 344 MW and relies mainly on high-speed diesel

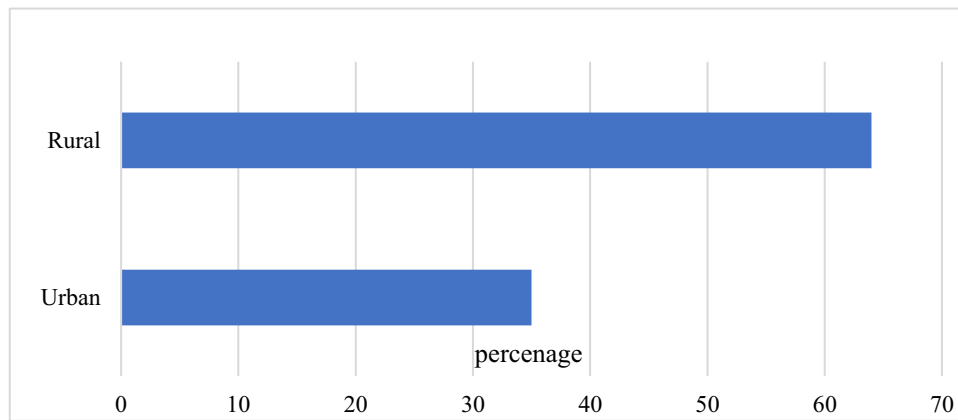


FIGURE 3 | Population distribution of Somalia [30].

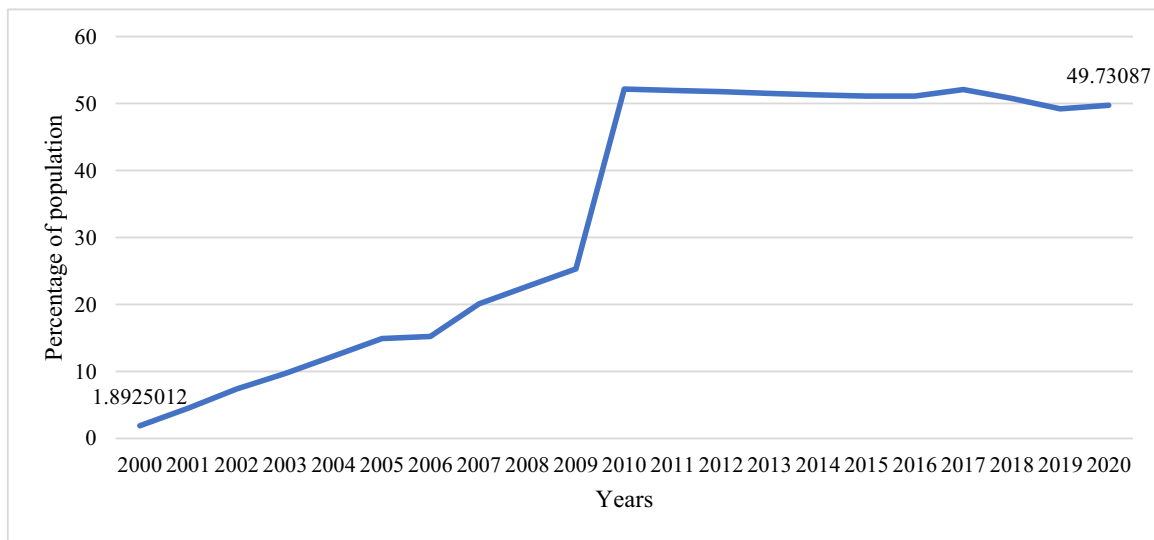


FIGURE 4 | Access to electricity to Somalia (% of population from 2000 to 2020) [31].

TABLE 2 | Population and GDP in (Millions) with growth rate from 2019 to 2022.

Years	Population	Population growth rate %	GDP	GDP growth rate %	House holds
2019	15.44	2.90	5,973	2.7%	2.62
2020	15.89	2.92	5,956	−0.3%	2.69
2021	16.36	2.93	6,131	2.9%	2.77
2022	16.84	2.95	6,250	1.9%	2.85

generator sets for electricity generation, using 121,000 L of diesel daily. This is expected to increase to 694,000 L by 2024 due to rapid urbanisation [1]. On the other hand Somalia has the high energy resource potential compared to any other country in Africa as recent research found did by the African Development Bank, it states that for onshore wind power, with a potential output of 30,000 to 45,000 MW. A surplus of 2000 kWh/m² might be produced from solar energy [31]. The country also has 200 billion cubic feet of proven gas reserves. Thus far, Somalia has failed to tap its conventional and nonconventional energy resources and resulting in a dependence on imports of these resources to fulfil its energy. The Somali Federal Government predicts that by 2027, the nation's total power generation capability will hit 700 MW. Through its “transformation scenario,” it plans to elevate this

capacity to 4600 MW by 2037 [3]. This scenario necessitates significant investment in the energy sector, efficient fund allocation at federal and state levels, and infrastructure development. Ministry of Energy and Water Resources (MEWR) officials cautiously support the goal but express concerns over current investment trends. The 2020 federal budget allocated no resources beyond operational costs for MEWR and other state bodies. Somalia's ninth national plan (2020–2024) outlines a strategic framework by the Ministry of Planning to advance renewable and nonrenewable energy sources. Proposed interventions include legislation for a national regulatory authority by 2021 and boosting electrification from 15% to 45% by 2024, mainly through foreign investment and other measures [35]. The Ministry of Energy and Water Resources in Somalia has initiated the Somalia Electricity Sector Recovery

TABLE 3 | Summary of the energy production of Somalia, Kenya, and the rest of Africa in 2020 [1].

Source	Total electricity production in Somalia (%)	Total electricity production in Kenya (%)	Total electricity production in Africa (%)
Fossil fuels	87.8	32.5	82.44
Solar	11.9	1	0.99
Wind	0.3	0.4	2.23
Hydropower	0.0	33.9	11.92
Nuclear	0.0	19.0	1.68
Geothermal	0.0	13.2	0.73

Project, backed by financial and technical aid from the World Bank. Project implementation will start shortly, aiming to enhance electricity accessibility across Somalia and develop energy infrastructure and supply sectors in Somalia [36]. The share of fossil fuels in the electricity generation mix was 87.8% in 2020, while renewable sources accounted for only 11.3%. The renewable share is much lower compared to that of Kenya and whole Africa as Table 3 depicts.

The Somali National Development Plan (NDP) Office indicates in its report that electricity generation, as depicted in, is continuously increasing, excluding independently generated electricity. Renewable energy (RE) has a total installed capacity of 41 MW (11.90%) for solar and 1 MW (0.3%) for wind. The majority of ESPs intend to boost their investments in renewable energy by the conclusion of 2023. As electricity demand grows, so does the generation capacity within ESPs. Figures 5 and 6.

As discussed in Table 4, the federal member states of Somalia have different energy supply capacities, tariffs, and levels of renewable energy utilisation.

The irradiation against elevation information for the 18 regions in Somalia is presented in Figure 7.

2.2 | Somali Energy Regulatory Framework

Somalia's governmental institutions for energy oversight are emerging with evolving frameworks. The Federal Government's Ministry of Energy and Water Resources (MoEWR) and Somaliland's Ministry of Energy and Minerals (MoEM) manage the electricity and energy sectors, though functionality varies. Absence of regulations jeopardises safety, quality, and consumer interests. Limited capacity for enforcement exacerbates challenges, despite initial policy efforts: (i) Creating and implementing a sector development blueprint, such as the Somali Power Sector Master Plan (PSMP), is aimed at laying the groundwork for establishing a contemporary energy sector in Somalia. This includes both putting the essential elements in place and passing necessary legislation such as The Electricity Act, along with the FGS Electricity Bill and Energy Policy, which received approval from the Council of Ministers in December 2020. Subsequently, the Electricity Act was endorsed by Parliament and signed into law by the federal president of Somalia in 2023 [37]. According to the bill, the Ministry of Energy and Water Resources is responsible for formulating policies and strategic planning, establishing legal frameworks,

preparing budgets, and managing key aspects of Somalia's economy in accordance with the laws. Additionally, the ministry represents Somalia in international organisations concerning negotiating agreements and treaties aligned with its frameworks, developing, and implementing regulations related to energy issues while considering all stakeholders, and engaging with and incorporating relevant international agreements into the country's energy policies. This allows the ministry to help shape Somalia's energy sector both domestically and globally in a sustainable manner that benefits the country's economic and political interests.

3 | Methodology and Data Description

This study employs the LEAP model to project Somalia's electricity demand for the period 2022–2050 and analyse energy supply pathways, considering various scenarios. LEAP, a widely used energy modelling tool, assesses energy policies and climate change mitigation. It enables scenario-based modelling for resource extraction, production, and consumption across sectors, making it particularly appropriate for data-scarce developing contexts such as Somalia due to its flexible structure and built-in global databases.

3.1 | Choice of a Framework

Energy systems analysis employs diverse tools based on research focus and subject. Selecting a tool entail assessing purpose (e.g., scenario building, forecasting or policy evaluation), spatiotemporal resolution, analytical approach (top-down, bottom-up), methodology (simulation, optimisation), and outputs [38]. Among these, LEAP was selected because it integrates both demand- and supply-side analyses and allows long-term scenario development without requiring highly complex datasets. Its transparency and adaptability make it effective for national-level planning in developing economies.

3.2 | Long-Range Energy Alternatives Planning System (LEAP)

The Long-range Energy Alternative Planning (LEAP) system is a soft-ware package developed by the Stockholm Environment Institute for energy modelling and analyses on climate change and energy policy [39]. Leap is a demand driving analysis software and its model features can be suitably determined by



FIGURE 5 | Location of Somalia in the African continent and the world.

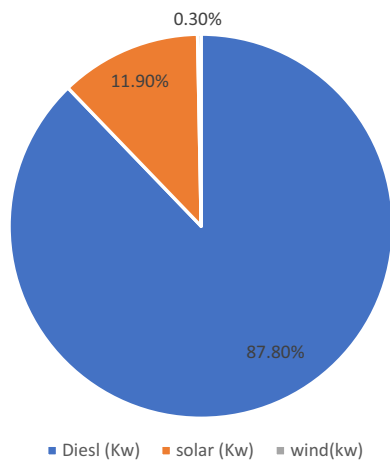


FIGURE 6 | Electricity production by type in Somalia (2020).

the data i.e. available to the user, the framework can be en-joined, the form of a hierarchical tree structure, of four different levels like sector, its subsector, device and end-use. The energy required by any sector is equivalent to the product of activity level and intensity of energy usage [40].

The LEAP model consists of six modules called, resources, conversion, demand prediction, end-use demand, impact of environment and analysis of cost-benefits. These modules collectively simulate future energy demand and supply, analyse system transformation, and estimate resulting emissions and costs.

LEAP can also forecast the environmental impacts of energy programs and evaluate economic feasibility based on resource availability, technology costs, and emission data [41]. The researchers using the LEAP system differ in terms of sectoral coverage, geographical scale, and research focus. The analysis

may be conducted at the national, regional, or multinational level, focusing on greenhouse gas emissions and energy supply. Sectoral analyses are also possible, focusing on cost assessment of the suggested scenarios or comparative evaluation of changes in external costs [18]. This diversity demonstrates its robustness for comparative analysis and policy evaluation.

TABLE 4 | Summary of Somalia's Energy supply with generating capacity and tariffs 2020 [1].

Regions FMS	Generating capacity (MW)		Tariff (\$/kWh)
	Diesel (MW)	RE (MW)	
Banadir Administration	124.9	9.5	0.36
Somaliland region	84	29	0.7
Puntland administration	52.75	2.5	1.00
Southwest administration	17.8	0	0.70
Hir-shabelle	9.6	1	1.00
Juba-land	8.4	0	0.90
Gal-mudug	48.8	0	1.00
Total	302.35	42	0.81

LEAP is a better option for this study due to its suitability for modelling and examining energy systems in areas with low and limited data availability. Considering the lack of historic time series data for some areas, LEAP may be a better choice because of its flexible structure and less stringent initial data requirements. LEAP is also a suitable modelling tool for conducting regional and national analysis [31]. While LEAP offers flexibility, a minimum dataset comprising demographic, economic, and sectoral energy statistics is required to build a base-year model. These data were collected and calibrated for Somalia using available national and international sources. However, the level of detail can be expanded based on the scope and goals of the specific modelling project [34].

In this study, the overall research procedure was organised into a structured sequence of activities designed to ensure systematic and transparent analysis, as illustrated in Figure 8. At the start of the study, the required electricity demand for Somalia was forecasted based on demographic, economic, and sectoral growth indicators. Following this, distribution losses and electricity generation profiles were modelled enabling the estimation of total supply requirements. Finally, the capacity of different technologies was calculated to meet the projected demand under each scenario, as presented in Figure 8.

The data used this study were collected from various credible sources to ensure accuracy and reliability. These included international scientific databases, official documents from the

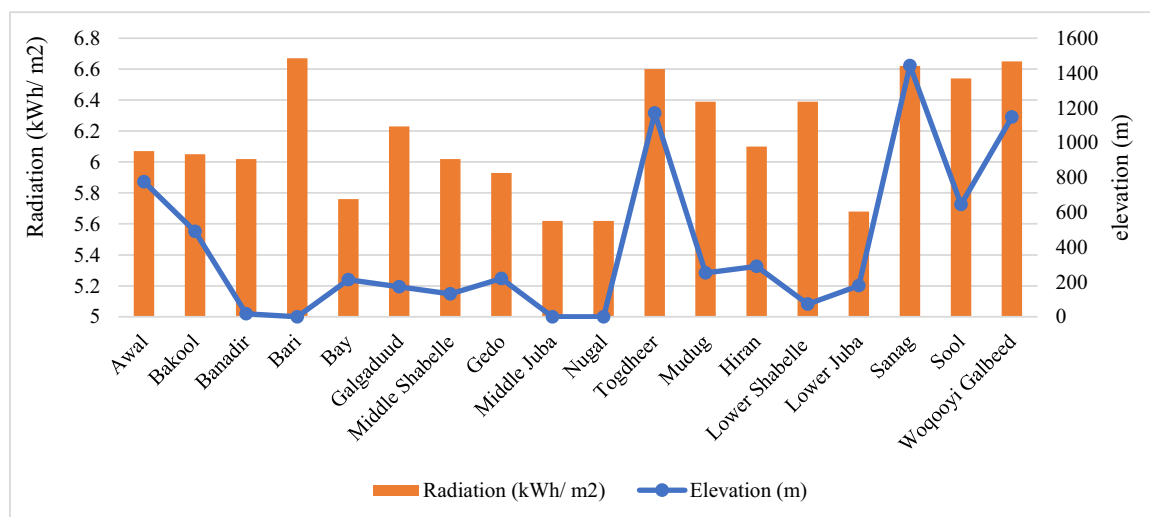


FIGURE 7 | Summary of the solar radiation data obtained For 18 Somalia Regions (2010–2020).

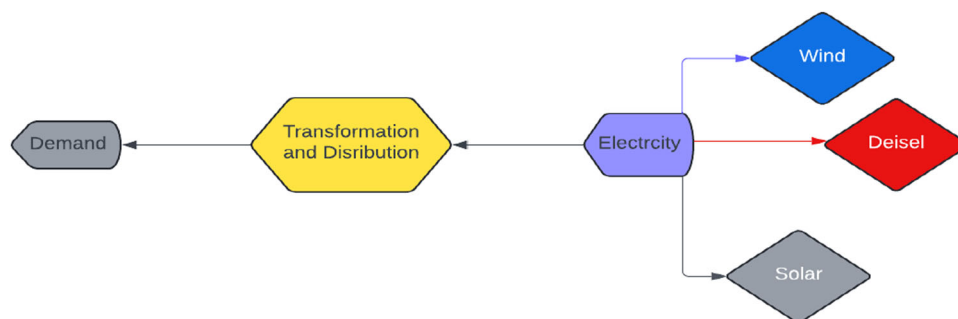


FIGURE 8 | Leap model for this study.

TABLE 5 | Various sources used to get information and data.

Sr#	Data source	Description
1	International Renewable Energy Agency	It's an intergovernmental organisation charged with fostering cooperation, advancing knowledge, and encouraging the adoption and sustainable utilisation of renewable energy [42]
2	US Agency for International Development	It's an autonomous entity within the United States government tasked mainly with managing civilian foreign aid and development assistance [43]
3	Banadir Electric Company (BECO)	This company is an electricity provider situated in Somalia, particularly in the capital city of Mogadishu. They deliver electricity services to residents, businesses, and industrial establishments in the area [44]
4	EU Global Technical Assistance Facility on Sustainable Energy	It's a centre that offers technical aid and capacity-building assistance to developing nations, aiding their shift toward sustainable energy systems [45]
5	Renewables Global Status Report	The Renewables Global Status Report, issued annually by REN21 (Renewable Energy Policy Network for the 21st Century), offers a thorough summary of the worldwide renewable energy market, industry, and policy environment [46]
6	World Bank Electricity Access	The World Bank endeavours to enhance electricity accessibility in developing nations through diverse projects and endeavours. They offer financial support and technical guidance aimed at enhancing electricity infrastructure, boosting energy efficiency, and advocating for renewable energy options [47]

Note: These diverse data sources, ranging from international organisations to reports and companies, provide valuable report about Somalia and worldwide.

TABLE 6 | Key assumptions.

Key assumptions	Input parameters	Growth %	Reference
Population	16.8 million	2.92	research paper
GDP	9.2 billion USD	3.5	World bank
Households	2.7million	2.6	NDPLAN 2020 to 2024
Access to electricity rate	6 million	6	NDPLAN 2020 to 2024

MoEWR and the Ministry of Natural Resources, and verified online repositories of regional, local, and international organisations. The compiled datasets are summarised in Table 5.

Overall, the methodological framework ensured consistency between data inputs, model assumptions, and scenario design, thereby enhancing the robustness of the energy-demand projections and policy implications derived from the LEAP model.

4 | Scenario Development

During the planning of electricity production and consumption, the supply-side scenarios showcase different pathways for the future. These pathways could provide policymakers with useful insights into the system effects while pursuing them. The scenarios also provide detailed information about resource usage, cost parameters, and greenhouse gas emissions under each scenario. This technique facilitates understanding the negative and positive impacts of individual policies. Furthermore, it allows for the assessment of how various pathways interact under policy considerations in various time horizons. In summary, the key functions of scenario development are to explain.

- Plausible events in the future, while bearing in mind the possible alternative results.

- Assessment of energy systems under various assumptions about the economy
- Planning through modelling techniques and checking the interactions between economic factors, emissions, population, and energy resources.

A brief overview of the key assumptions with its input parameters and growth rate of this study are presented in Table 6.

This study has created three supply side scenarios to align with the forecasted demand from 2020 to 2050. Table 7 provides a concise overview of the four scenarios. Further details regarding each scenario are provided below.

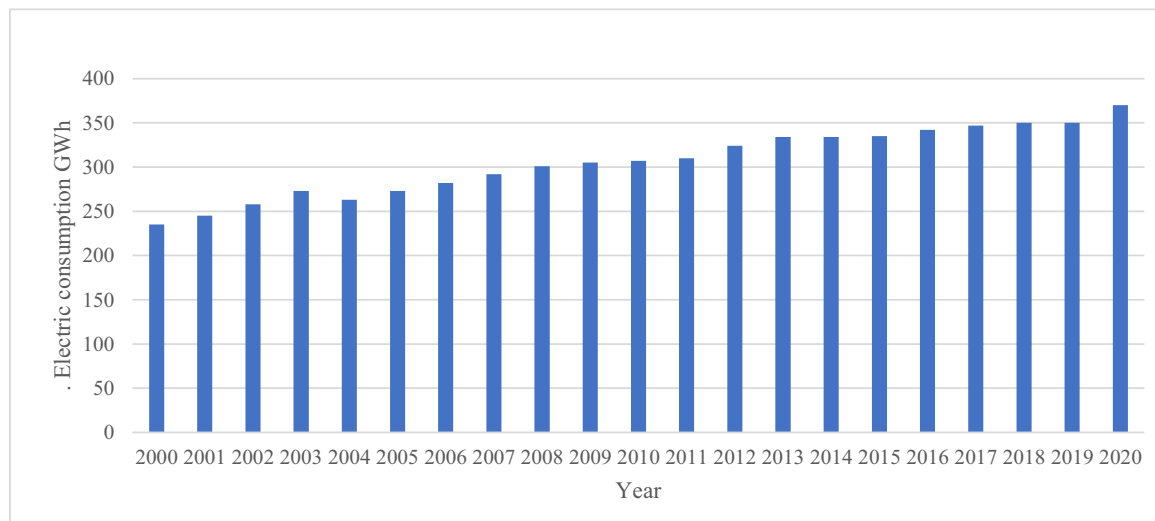
As Figure 9 illustrates the electricity Consumption of Somalia from 2000 to 2020, this historical data will help the model creation and scenario development.

5 | Results

This subpart of the paper presents the outcomes of this study, focusing on forecasting both demand and supply-sides, along with related techno-economic findings and CO₂ emissions.

TABLE 7 | Scenario alternatives of this study.

Indicators	BAU(REF)	RET	Visionary transition
Share of Renewable energy	11.3	70	100
Energy options	In this scenario, the current plan and policy of the government are implemented	In this scenario, renewable energy sources and technologies are prioritised.	Advanced use of renewable sources.
Energy restrictions	No restrictions on fossil fuel	Moderate restriction on fossil fuel	High restriction on fossil fuel
Energy alternative promotion	No promotion	Small promotions	High/advance promotions

**FIGURE 9** | Electricity consumption of Somalia for the period 2000-2020.

5.1 | Demand Forecast

The Somali electricity demand forecast from 2020 to 2050, under three different scenarios, are shown in Figure 10 the growth related to electricity demand estimated in this result attribute the increase in population, rise of the population who have accesses to the electricity, swift expansion in urbanisation, improved living standards, and GDP growth. In all estimated electricity demand of Somalia in 2050 is projected to be 1814.2 GWh which was only 133.2 GWh in 2020, and this growth approximately 13.62 times higher, to be base year, and the evaluated annual 1.88% is projected. Since REF, RET, and VT scenarios are driven by similar demand-side assumptions, the demand forecast for these three scenarios is the same: 1814.2. If we compare the demand projected by the Kenyan government, which is 48,514 GWh in 2037 using LEAP [22], we can see a significant difference in energy consumption between Kenya and Somalia.

The forecast shows an increase in electricity demand due to population growth and changes in lifestyle. Additionally, economic growth and an increase in the population with access to electricity may also contribute to this increase.

5.2 | Supply Side Scenario

The electricity SupplySide estimation of every of three scenarios (BAU, RET, VT) of this study contains of the electric production to fulfil the required demand, the installed volume to make sure the required production, the cost analysis in regard of Net Present Value and finally the estimation CO₂ emissions.

5.2.1 | Generation

The model projects and estimates electricity production in the study period across different scenarios by employing a variety of fuel and technology combinations to satisfy the required demand, Figure 9 presents the electricity production under the three scenarios of this study period.

In the business-as-usual scenario the percentage of diesel in electricity generation mix is 87% in 2020. Which will be 79.1% at 2050 according to the Somalia's aims to movement towards a more sustainable and less diesel-dependent electricity production system strategies, leveraging its abundant solar and wind resources. The share of solar, and wind in the business-as-usual year is about 10%, and 3% respectively which is estimated to

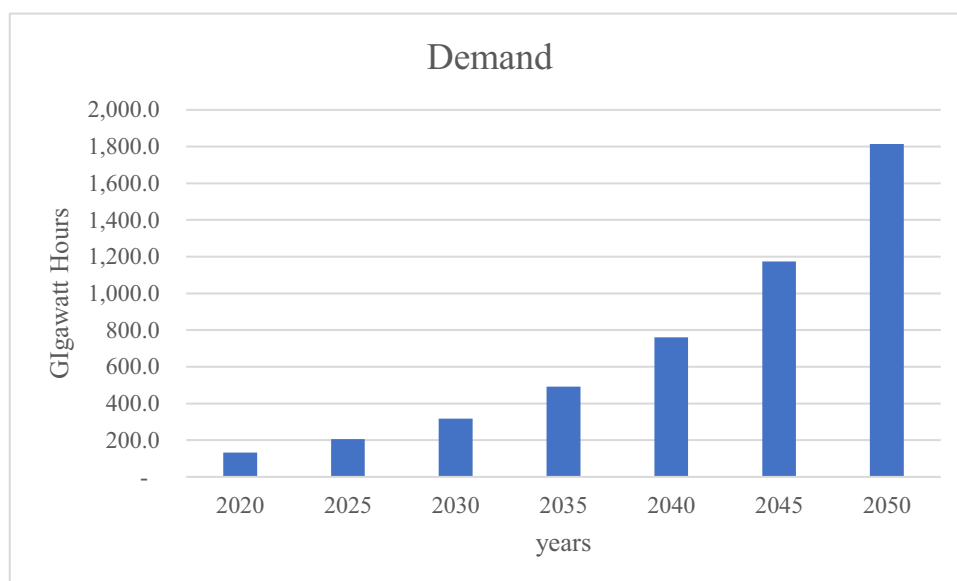


FIGURE 10 | Forecasted electricity demand (GWh).

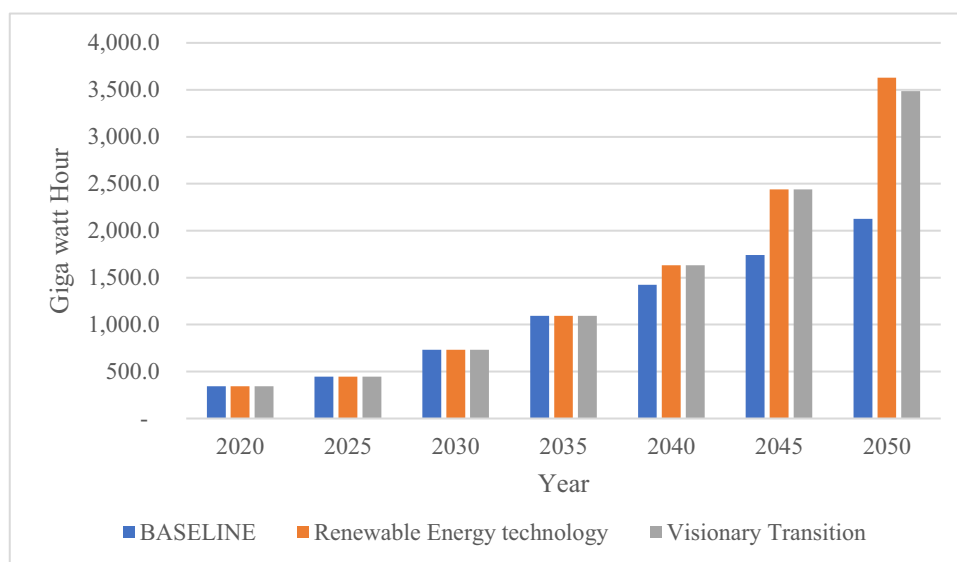


FIGURE 11 | Installed volume in all scenarios in study time.

be 20.4%, 0.5%, and respectively in 2050. In the Renewable Energy Transition (RET) scenario, it is projected that by 2050, solar, and wind, will contribute 45%, and 25%, respectively to the energy mix. Overall, renewable energy is projected to increase from 13% in 2020 to 70.3% in 2050 in this scenario. In the Visionary Transition Scenario with no production of electricity from diesel 100% renewable electricity supply by 2050. This indicates a comprehensive expansion of renewable energy across all sectors and areas as depicted in Figure 10. In this VT scenario, it's anticipated that solar energy will constitute approximately 68.6% of the total share, with wind energy accounting for the remaining 31.5%. Figure 11 in the renewable energy technology scenario, there is moderate usage of renewable sources. The study assumes that the Somali government will transition more than 70% of its electricity generation to clean methods. In the visionary transition scenario, there is advanced usage of renewable sources. The study assumes that the Somali government will transition all its electricity generation to clean methods. Figure 12.

5.2.2 | Installed Capacity

The installed volume of diverse power plant kind has been calculated to meet the requisite electricity production demands across the three distinct scenarios.

5.2.2.1 | Business as Usual (BUA). The predominant installed capacities for electricity generation are projected primarily from Speed Diesel Generators, accounting for 303.2 GW, or 87.2%, of the total capacity in the base-year. This figure is optimistically forecast to be 1680.5 GW, representing 79.1% of the overall capacity, by the year 2050. At the same time, a significant augmentation in the proportion of renewable energy sources to the national electricity mix is anticipated. Specifically, it is estimated that solar and wind energy will collectively contribute approximately 445.1 GW, or 20.9%, of the country's total electricity generation capacity by 2050, a substantial

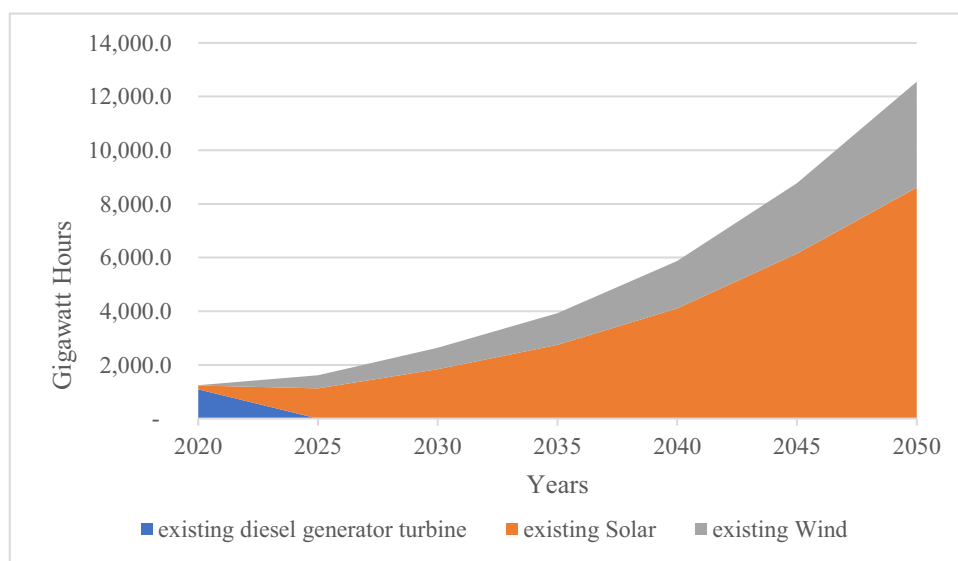


FIGURE 12 | Visionary transition scenario.

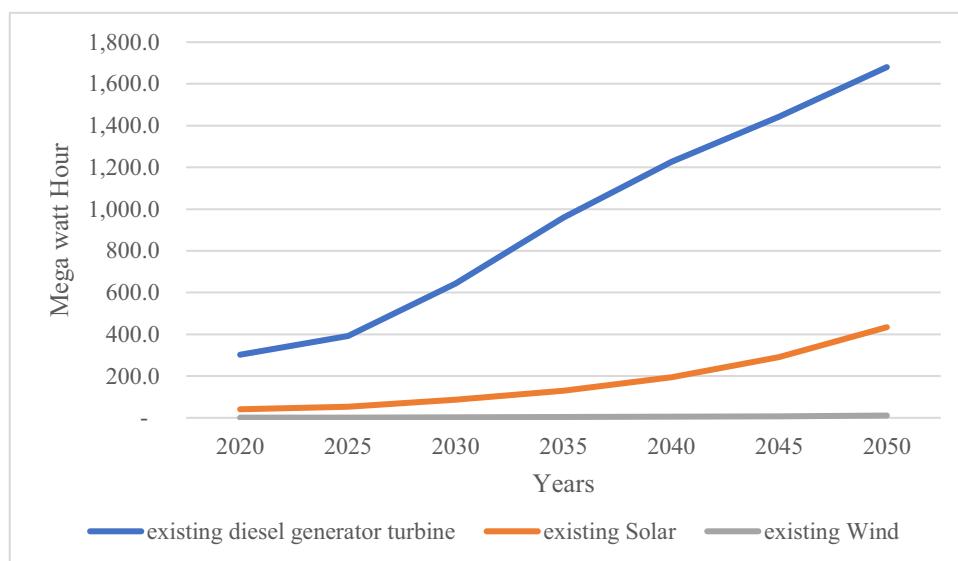


FIGURE 13 | The overall capacity in BAU, by the year 2050.

increase from a mere 42 GW, or 11.3%, in the base year this projection it aligns to the Somali government strategies as depicted in Figure 13.

5.2.2.2 | Renewable Technology (RET). In relation to the RET scenario, the assessed capacity for wind and solar energy is onshore wind power, with a potential output of 30,000 to 45,000 MW. A surplus of 2000 kWh/m² might be produced from solar energy, according to this scenario. The initial installed capacity of solar and wind, standing at 41 MW (11.9%), 1 MW (0.3%) respectively in the baseline year, is projected to experience significant growth to 1641 GW (42.2%) and 911.3 GW (25.1%). Respectively at the end of 2050, the electricity generation from diesel reduces to 1078.3 GW, representing 29.7% of the total capacity. The total proportion of renewable energy within the installed capacity is forecasted to reach 70.3% by the year 2050, as shown in Figures 13 and 14.

5.2.2.3 | Visionary Transition Scenario (VST). In the VTS scenario, the utilisation of fossil fuels for electricity generation has been phased out by the year 2050. The breakdown of electricity generation for the year 2050 is as follows: 2392 GW (68.6%) from solar energy and 1095 GW (31.3%) from wind power, as shown in Figure 15.

5.2.3 | CO₂ Emissions

Achieving carbon neutrality by 2050 is crucial for Somalia to uphold its obligations outlined in the Paris Agreement and to play a significant role in mitigating global warming as Somalia is ranked among the most vulnerable countries to climate change in the world, despite its overall carbon emissions being only 0.8 percent. According to this study the base in 2020 the carbon emission from electric generation in the country was 312 thousand metric tonnes and this number

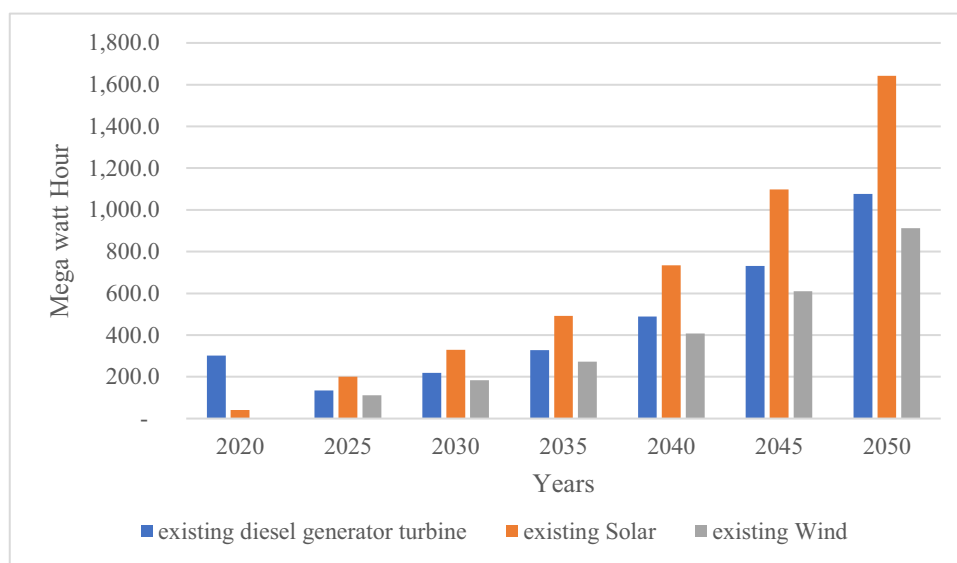


FIGURE 14 | The overall capacity in RET, by the year 2050.

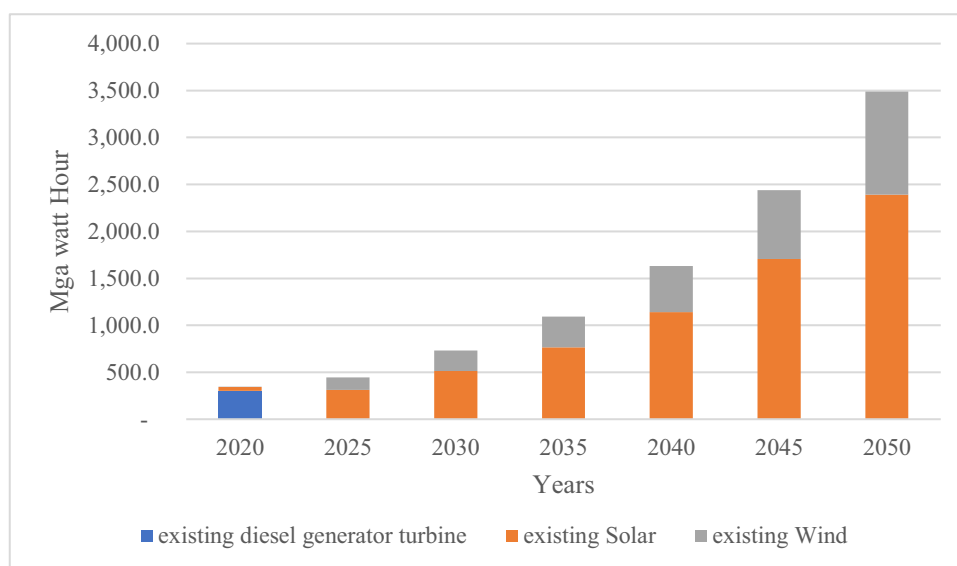


FIGURE 15 | VTS scenario full capacity.

has been estimated to reach 2194.6 thousand metric tonnes in 2050, under the Business-as-usual scenario. Figure 16 demonstrates CO₂ emissions from electric generation under the BAU and other scenarios in this study. In the Renewable Energy Transition (RET) scenario, where 70.3% of electricity generation comes from renewable sources, carbon dioxide (CO₂) emissions are forecasted to be significantly lower compared to other BAU in the year 2050 are 1406.3 thousand metric tonnes which is 1.56 times smaller than the BAU scenario for the same time. The CO₂ emissions in VTS scenario are estimated to be NetZero in 2050 since the reduction in fossil fuel electricity generation and complete usage of renewable energy to produce electricity. The greenhouse gas (GHG) emissions assessed for each generation scenario are based on the IPCC Tier 1 database, which refers to a methodology for estimating greenhouse gas emissions and

removals developed by the IPCC Tier 1 and is the simplest and least data-intensive approach.

5.2.4 | Cost Evaluation

In this study, the overall cost of the system is represented by NPV, comprising capital expenses, fixed operational and maintenance (O&M) costs, these costs are discounted at rates of 5%, over the study time for each scenario outlined as depicted in Figure 17. The VST scenario exhibits the shortest NPV compared to BAU, RET, scenarios in the discount rates. This discount is close to the State Bank of Somalia policy rate of 4.95%, the VST scenario demonstrates the lowest NPV due to the transition away from fossil fuels toward solar and wind energy usage, coupled with a complete pivot from reliance on imported fossil fuels, requires several key

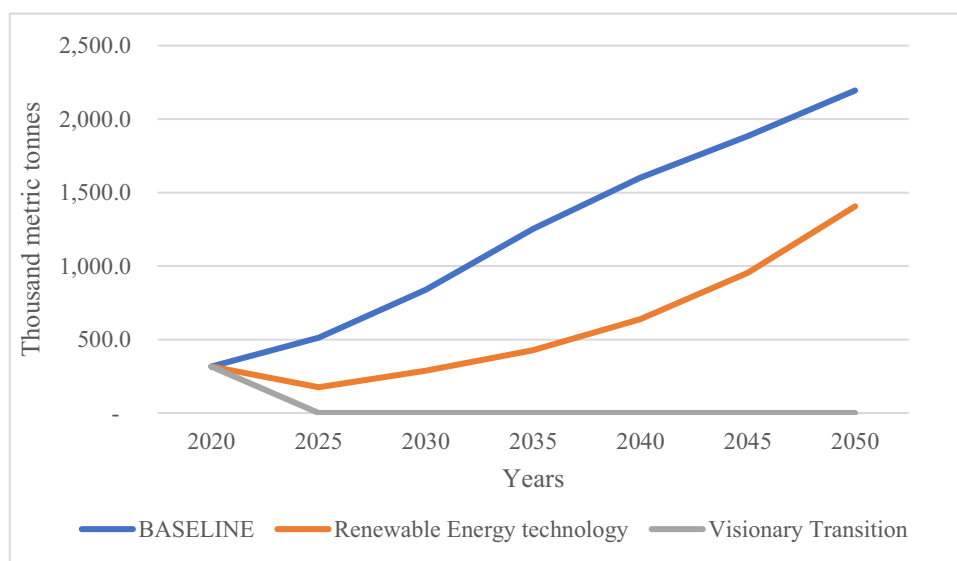


FIGURE 16 | CO₂ emissions in all three scenarios from 2020 to 2050.

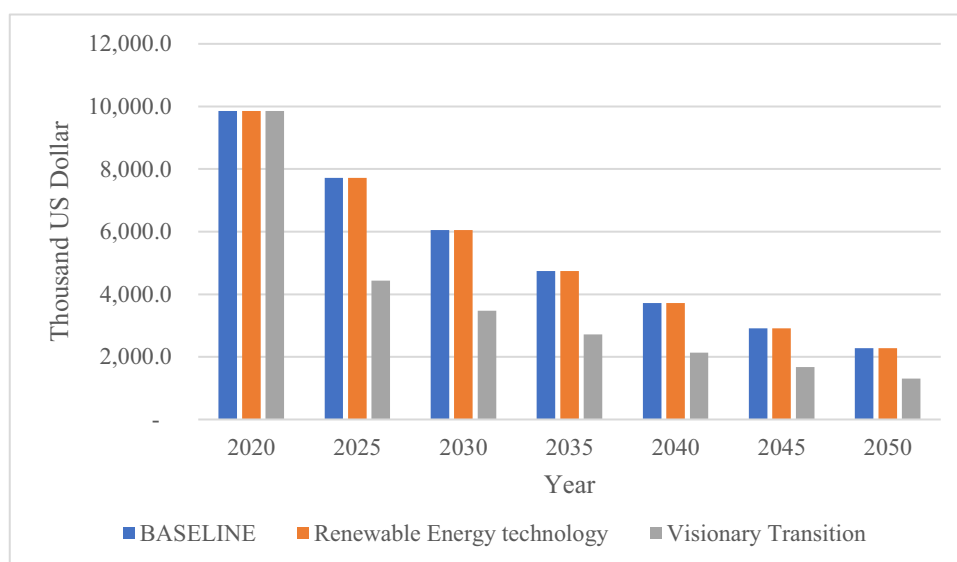


FIGURE 17 | Net previous value at different scenario.

actions, other scenarios indicate negative NPV which demonstrate unprofitable in all the study period.

6 | Analysis/Discussion

This study employs the LEAP model to project Somalia's electricity demand (2022–2050) and analyse energy supply pathways, considering three scenarios. The LEAP framework enables scenario-based modelling of demand, supply, and environmental outcomes, allowing evaluation of policy options in data-scarce environments like Somalia.

The study's results underscore the pressing need for Somalia to address its growing electricity demand while simultaneously reducing CO₂ emissions to combat climate change. Electricity demand is projected to increase more than twelvefold by 2050, driven by population growth, GDP expansion, and rising

electrification. The projected growth (from 133.2 GWh in 2020 to 1814.2 GWh in 2050) is large. Similar growth patterns have been reported by Mirjat et al. [11] for Pakistan, confirming that socioeconomic progress in developing countries strongly influences long-term demand forecasts. However, unlike Pakistan or Nepal, Somalia lacks a unified grid or reliable baseline statistics. By constructing an integrated dataset and simulating combined Renewable Transition and Visionary Transition scenarios, this study introduces new empirical evidence on how decentralised renewables could function in fragile, low-data environments. Therefore, meeting future demand through sustainable energy sources becomes vital to avoid replicating the fossil-fuel dependence seen in neighbouring systems.

The switch towards renewable technology sources, particularly solar and wind, is crucial for achieving carbon neutrality by 2050 and aligning with global climate goals outlined in the Paris Agreement. Comparable studies, such as Hu et al. [13] in Beijing, demonstrated that integrated energy-efficiency and renewable

policies can significantly reduce emissions while sustaining economic growth. Likewise, Javid et al. [16] highlighted the role of decentralised clean-energy networks in promoting inclusive economic development, an approach that Somalia could adopt to expand rural electrification and resilience.

The findings also emphasise the economic benefits of transitioning towards renewable energy, as evidenced by the lower NPV in the VST scenario compared to BAU and RET scenarios. This suggests that investing in renewable energy technologies cannot solely lower carbon emissions but also leads the way to cost savings in the long-term. This outcome aligns with Mirjat et al. [11], who found that renewable-rich scenarios yield lower system costs and emissions for Pakistan's power sector, and with Poudyal et al. [21], who demonstrated that replacing imported fossil fuels with indigenous renewables improves both affordability and energy security. These consistent findings reinforce the economic rationale for Somalia's transition.

By phasing out fossil fuel-based electricity generation and embracing renewable energy alternatives, Somalia can place itself as a leader in sustainable energy practices while contributing to global efforts to combat climate change. Furthermore, this transition would support the Sustainable Development Goals by reducing environmental degradation and strengthening climate resilience, as discussed in the sustainability literature [8].

Overall, the study's results provide a comprehensive roadmap for Somalia to navigate the challenges of increasing electricity demand, reduce CO₂ emissions, and achieve sustainable economic growth through the adoption of renewable energy technologies. By prioritising renewable energy investments and implementing effective policy measures, Somalia can pave the way for a greener and more resilient energy future.

7 | Conclusion and Policy Recommendations

This study provides the first integrated, scenario-based LEAP assessment for Somalia's electricity sector, linking technological, economic, and environmental factors. It offers new quantitative benchmarks for demand growth, renewable potential, and emission reduction in a country where no such analysis previously existed. The results provide a clear, internally coherent, and plausible depiction of the country's potential energy future. The model outcomes illustrate that energy planning in Somalia must simultaneously consider growing demand, cost efficiency, and emission reduction.

The three simulated scenarios (BAU, RET, and VTS) represent alternative policy pathways for Somalia power sector showing how varying levels of renewable integration and fossil-fuel dependency could shape the nation's electricity mix, installed capacity, emissions, and cost structures.

In all estimated electricity demand of Somalia in 2050 is projected to be 1814.2 GWH which was only 133.2 GWH in 2020 and this growth, approximately 13.62 times higher, to be base year, the evaluated annual growth 1.88% is projected. All supply-side scenarios were devised to fulfil the electricity demand forecasted. To satisfy this growing demand, the

analysis reveals that a hybrid strategy combining features of the Renewable Energy Transition (RET) and Visionary Transition Scenario (VTS) would provide the optimal balance between economic feasibility and environmental sustainability.

According to the findings of this study, Somalia's current policy trajectory is insufficient to achieve energy independence or sustainability; continued reliance on imported fuels will perpetuate high costs and emissions. However, adopting a structured renewable-energy transition can substantially reduce fossil-fuel imports, improve energy security, and stabilise long-term system costs. In this context, it's crucial to employ energy modelling at both governmental and academic levels for analysing energy systems and formulating policies. This allows testing of various hypotheses and policy perspectives. Furthermore, future policy design should integrate sustainable financing mechanisms, such as green bonds, climate funds, and blended finance models to mobilise investment in renewable infrastructure. As highlighted by sustainable finance studies [48], access to climate-oriented funding can accelerate low-carbon transitions, particularly in developing and fragile economies like Somalia.

However, the findings from energy modelling should be interpreted as policy-support tools rather than prescriptions, helping decision-makers explore feasible pathways for an affordable, secure, and decarbonised energy future.

- Streamline and strengthen institutional frameworks for the power and energy sector, supported by targeted capacity-building and research programs. Effective policy design and implementation are essential to realising alternative energy pathways. This includes investing in research and development for renewable energy, addressing administrative barriers and fostering public awareness for energy conservation and efficiency. Future work could refine these findings by coupling LEAP with optimisation or sustainable-finance modules to assess investment feasibility and cross-border energy-trade options.
- Introduce carbon pricing instruments, such as a carbon tax or cap-and-trade system to incentivize the adoption of low-carbon technologies and internalise environmental costs.
- Align national energy policies with climate strategies mandated under the Paris Agreement, by scaling up renewable deployment, particularly solar and wind through public-private partnerships, concessional financing, and technology transfer. This approach will not only enhance energy access but also support economic growth and environmental protection.

Implementing the above recommendations and embarking on a comprehensive national energy plan for power policy formulation are expected to lead to programs that enhance Somalia's energy security, reduce dependency on imported fuels, and advance the nation toward a sustainable and low-carbon energy future.

Author Contributions

Abdisalan Moallim Hirsi: conceptualisation, data curation, formal analysis, investigation, methodology, resources, software, validation, writing – original draft, writing – review and editing. **Shoaib Ahmed**

Khatrri: conceptualisation, data curation, methodology, resources, writing – original draft, writing – review and editing. **Nayyar Hussain Mirjat:** conceptualisation, formal analysis, methodology, software, writing – original draft. **Musavir Hussain:** data curation, formal analysis, methodology, writing – original draft, writing – review and editing. **Laveet Kumar:** formal analysis, investigation, methodology, supervision, validation, visualisation, writing – review and editing. **MD Shouquat Hossain:** data curation, formal analysis, software, visualisation, writing – review and editing.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. A. M. Samatar, S. Mekhilef, H. Mokhlis, et al., “The Utilization and Potential of Solar Energy in Somalia: Current State and Prospects,” *Energy Strategy Reviews* 48 (2023): 101108.
2. F. G. o. Somalia., “Somalia’s Intended Nationally Determined Contributions (INDCs),” 2015.
3. I. a. E. D. Ministry of Planning, Federal Government of Somalia., “Somalia National Development Plan 2020-2024: The Path to a Just, Stable and Prosperous Somalia,” 2020.
4. P. Giganti, C. Barra, and P. M. Falcone, “Unveiling the Nexus of Institutional Quality and Renewable Energy Utilizing a Topic Modelling Approach,” *Renewable and Sustainable Energy Reviews* 214 (2025): 115516.
5. P. Giganti and P. M. Falcone, “Unveiling Latent Topics in the Interplay of Circular Economy and Energy Transition: A Topic Modelling Approach,” *Resources, Conservation And Recycling* 219 (2025): 108318.
6. P. M. Falcone and G. Errichiello, “Energy Justice in the 21st Century: Policy Interventions and Advocacy for a Fair Energy Transition,” *AIMS Environmental Science* 12 (2025): 594–614.
7. C. Barra and P. M. Falcone, “Unraveling the Impact of Economic Policy Uncertainty on Environmental Efficiency: How Do Institutional Quality and Political Orientation Matter?,” *Economics & Politics* 36 (2024): 1450–1490.
8. P. M. Falcone and I. Tutore, “Mapping the Nexus: A Bibliometric Analysis and Social Network Analysis of Transformative Innovation Policies and Sustainable Development Goals,” *Business Strategy and the Environment* 34 (2025): 2423–2435.
9. L. Aldieri, C. Barra, P. M. Falcone, and C. P. Vinci, “Socio-Political Determinants of Circular Economy Behavior: A Cross-Sectional Analysis Across Italy,” *Socio-Economic Planning Sciences* 100 (2025): 102252.
10. M. Davis, M. Ahiduzzaman, and A. Kumar, “How to Model a Complex National Energy System? Developing an Integrated Energy Systems Framework for Long-Term Energy and Emissions Analysis,” *International Journal of Global Warming* 17 (2019): 23–58.
11. N. H. Mirjat, M. A. Uqaili, K. Harijan, G. D. Walasai, M. A. H. Mondal, and H. Sahin, “Long-Term Electricity Demand Forecast and Supply Side Scenarios for Pakistan (2015–2050): A Leap Model Application for Policy Analysis,” *Energy* 165 (2018): 512–526.
12. M. Asim, A. Qamar, A. Kanwal, et al., “Opportunities and Challenges for Renewable Energy Utilization in Pakistan,” *Sustainability* 14 (2022): 10947.
13. J. Hu, G. Liu, and F. Meng, “Estimates of The Effectiveness for Urban Energy Conservation and Carbon Abate-Ment Policies: The Case of Beijing City, China,” *Journal of Environmental Accounting and Management* 6 (2018): 199–214.
14. C. Su and F. Urban, “Circular Economy for Clean Energy Transitions: A New Opportunity Under the COVID-19 Pandemic,” *Applied Energy* 289 (2021): 116666.
15. D. Castrejón, A. M. Zavala, J. A. Flores, M. P. Flores, and D. Barrón, “Analysis of the Contribution of CCS to Achieve the Objectives of Mexico to Reduce Ghg Emissions,” *International Journal of Greenhouse Gas Control* 71 (2018): 184–193.
16. I. Javid, A. Chauhan, S. Thappa, et al., “Futuristic Decentralized Clean Energy Networks in View of Inclusive-Economic Growth and Sustainable Society,” *Journal of Cleaner Production* 309 (2021): 127304.
17. I. Ari, “A Low Carbon Pathway for the Turkish Electricity Generation Sector,” *Green and Low-Carbon Economy* 1 (2023): 147–153.
18. J. H. Hong, J. Kim, W. Son, et al., “Long-Term Energy Strategy Scenarios for South Korea: Transition to a Sustainable Energy System,” *Energy Policy* 127 (2019): 425–437.
19. S. Zhong, B. Su, X. Lin, and T. S. Ng, “Moving Towards a Net-Zero Emissions Economy: The Case of Indonesia,” *Available at SSRN* 4295163, 2022.
20. M. M. Hasan and W. Chongbo, “Estimating Energy-Related CO2 Emission Growth in Bangladesh: The Lmdi Decomposition Method Approach,” *Energy Strategy Reviews* 32 (2020): 100565.
21. R. Poudyal, P. Loskot, R. Nepal, R. Parajuli, and S. K. Khadka, “Mitigating the Current Energy Crisis in Nepal With Renewable Energy Sources,” *Renewable and Sustainable Energy Reviews* 116 (2019): 109388.
22. V. Wambui, F. Njoka, J. Muguthu, and P. Ndwali, “Scenario Analysis of Electricity Pathways in Kenya Using Low Emissions Analysis Platform and the Next Energy Modeling System for Optimization,” *Renewable and Sustainable Energy Reviews* 168 (2022): 112871.
23. S. Abrar and H. Farzaneh, “Scenario Analysis of the Low Emission Energy System in Pakistan Using Integrated Energy Demand-Supply Modeling Approach,” *Energies* 14 (2021): 3303.
24. F. M. de A. Collaco, S. G. Simoes, L. P. Dias, N. Duic, J. Seixas, and C. Bermann, “The Dawn of Urban Energy Planning–Synergies Between Energy and Urban Planning for São Paulo (Brazil) Megacity,” *Journal of Cleaner Production* 215 (2019): 458–479.
25. D. Icaza, D. Borge-Diez, and S. P. Galindo, “Proposal of 100% Renewable Energy Production for the City of Cuenca-Ecuador by 2050,” *Renewable Energy* 170 (2021): 1324–1341.
26. L. Malka, F. Bidaj, A. Kuriqi, A. Jaku, R. Roçi, and A. Gebremedhin, “Energy System Analysis With a Focus on Future Energy Demand Projections: The Case of Norway,” *Energy* 272 (2023): 127107.
27. M. T. Gebre, J. Hwang, and G. Biru, “Electricity Demand Analysis and Forecasting: The Case of Gada Special Economic Zone,” *Heliyon* 10 (2024): e25364.
28. H. A. Halarou, A. L. Bonkaney, D. Abdourahimoun, Y. Moumouni, and S. Madougou, “Application of a Long-Range Energy Alternatives Planning System to Assess Electricity Supply and Demand in Niger,” in *2024 International Conference on Green Energy, Computing and Sustainable Technology (GECOST)*, 2024, pp. 217–223.
29. P. Golfam, P.-S. Ashofteh, and H. A. Loáiciga, “Forecasting Long-Term Energy Demand and Reductions in GHG Emissions,” *Energy Efficiency* 17 (2024): 19.
30. C. Edwards, “Powering Ahead,” *Engineering & Technology* 8 (2013): 52–57, 12/01.
31. S.-S. Africa. (2025). *World Bank Group*.
32. F. R. o. Somalia., “National Environmental Policy,” 2019.
33. M. A. M. Omar., “Accord University Knowledge & Vision.”
34. W. contributors. (10 October). *Somalia*. <https://en.wikipedia.org/wiki/Somalia>.
35. F. G. o. S. Ministry of Energy and Water Resources., “Power Master Plan - Somalia,” 2018.

36. M. o. E. a. W. R. M. Federal Government of Somalia., "Grievance Redress Mechanism (GRM) Implementation Plan: Somalia Electricity Sector Recovery Project (SESERP)," 2023.
37. M. o. E. a. W. R. M. Federal Government of Somalia., "National Electricity Act (Somalia)/Hindisa-Sharciyeedka Korontada Qaranka," 2023.
38. G. P.-A. Colonel-Bertrand, "Modelling of the Hong Kong Power System by 2030," 2020.
39. W. G. Santika, M. Anisuzzaman, Y. Simsek, P. A. Bahri, G. M. Shafiullah, and T. Urmee, "Implications of the Sustainable Development Goals on National Energy Demand: The Case of Indonesia," *Energy* 196 (2020): 117100.
40. R. Chen, Z. Rao, and S. Liao, "Hybrid LEAP Modeling Method for Long-Term Energy Demand Forecasting of Regions With Limited Statistical Data," *Journal of Central South University* 26 (2019): 2136–2148.
41. D. Yang, D. Liu, A. Huang, J. Lin, and L. Xu, "Critical Transformation Pathways and Socio-Environmental Benefits of Energy Substitution Using a Leap Scenario Modeling," *Renewable and Sustainable Energy Reviews* 135 (2021): 110116.
42. U. U. N. F. C. o. C. Change. *Youth4Capacity-1: Youth-Led Acceleration of Just Energy Transition & Resilience in Africa*. <https://unfccc.int/event/youth4capacity-1-youth-led-acceleration-of-just-energy-transition-resilience-in-africa>.
43. W. contributors. *United States Agency for International Development*. https://en.wikipedia.org/wiki/United_States_Agency_for_International_Development?utm_source=chatgpt.com.
44. I. T. A. U.S. Department of Commerce. (2024). *Somalia – Energy and Electricity*. https://www.trade.gov/country-commercial-guides/somalia-energy-and-electricity?utm_source=chatgpt.com.
45. C. I. Funds. *Technical Assistance Facility*. <https://www.cif.org/technical-assistance-facility>.
46. R. Secretariat, "Renewables 2023 Global Status Report: Global Overview," 2023.
47. W. Bank. (2025). *Energy Overview: Development News, Research, Data*. <https://www.worldbank.org/en/topic/energy/overview>.
48. P. M. Falcone and E. Sica, *Sustainable Finance and the Global Health Crisis* (Taylor & Francis, 2023).